

NL BANKING

Customer Behaviour, Transaction Performance and Risk Analysis

Business Analytics Project Report

25,000	Customers
100,000	Transactions
₹164.25 Cr	Total transaction value
₹16,424.67	Average transaction amount
91.15%	Success rate
5.38%	Fraud rate

This report converts the completed Excel analysis into a clear business story: where customer engagement is strongest, where transaction value differs, where fraud exposure is concentrated, and what management should do next.

Prepared for project submission

1. Executive Summary

NL Banking has a large transaction base, but the data does not point to one universal driver of customer behaviour or risk. A few dimensions create clear separation, while others remain broadly similar across groups.

Headline findings

- **Card ownership is the clearest engagement signal.** Average transactions per customer rise from 3.07 for one card to 6.22 for two cards and 9.09 for three cards.
- **International activity is the clearest risk-value signal.** International transactions average ₹37,022.80 versus ₹15,333.03 domestically, while observed fraud is 19.15% versus 4.65%.
- **POS is the dominant transaction channel.** POS accounts for 48,341 of 100,000 transactions. Channel success rates remain close to 90.7%-91.2%.
- **Merchant category creates large value differences.** Electronics averages ₹37,333.23 per transaction, while Food Delivery averages ₹2,986.03.
- **Merchant risk labels do not separate fraud well.** Fraud rates are almost identical for High, Low and Medium risk merchants: 5.40%, 5.38% and 5.36%.

Executive conclusion

The practical focus should be selective rather than broad: deepen engagement where behaviour already shows stronger usage, protect and monitor high-value international activity, strengthen fraud detection beyond merchant risk labels, and keep high-volume channels reliable.

2. Business Context, Objective and Scope

Business context

The project evaluates a banking transaction dataset from the perspective of customer behaviour, transaction performance and risk. The business needs a simple view of what drives engagement, where transaction value differs, which transaction channels matter most, and where fraud exposure is concentrated.

Business objective

Identify the customer, transaction and merchant patterns that matter most for engagement, transaction value and risk, and translate those findings into practical actions and stakeholder-ready dashboards.

Core business questions

- Which customer behaviours are associated with stronger transaction engagement?
- Which customer or merchant dimensions create meaningful differences in transaction value?
- Which channels and payment methods carry the most operational volume?
- Where is observed fraud exposure concentrated, especially across international activity and merchant risk?
- Which findings are strong enough to justify a business action, and which dimensions should remain supporting context?

Scope and analytical principle

The analysis uses 100,000 transaction records and links transaction, customer, card and merchant information. Results are descriptive: observed differences are treated as associations and prioritisation signals, not as proof of causality. External benchmarks, profitability, operating costs and causal inference are outside the available project scope.

3. KPI and Metric Framework

Primary success measures

Metric	Definition	Business purpose
Transaction Volume	Count of transactions	Measure scale and usage by channel or method
Average Transaction Value	Average Transaction_Amount	Measure value per transaction
Transactions per Customer	Transaction count relative to customer count	Measure customer engagement frequency
Success Rate	Successful transactions / total transactions	Measure transaction reliability
Fraud Rate	Fraud transactions / total transactions	Measure observed risk exposure

Analytical dimensions

Customer: card ownership, customer segment, age and income. **Transaction:** channel, device, payment method and transaction type. **Merchant:** category, risk level and geography. **Risk:** fraud rate and international exposure.

KPI interpretation

The project intentionally separates scale metrics from value and risk metrics. High transaction volume does not automatically mean high value or high risk; the analysis compares these dimensions together before recommending action.

4. Data Preparation, Audit and Analytical Approach

Source structure

Source	Primary role
RAW - Transaction_data	Transaction ID, amount, channel, status, fraud, international flag and transaction-level fields
RAW - Customer_data	Customer attributes such as age, income, segment and geography
RAW - Cards_data	Card ownership and card-level characteristics
RAW - Merchant_table	Merchant category, risk level and merchant geography

Audit and traceability

The Excel workbook keeps raw data separate from analytical outputs and uses dedicated EDA, hypothesis validation and recommendation sheets. Tableau uses the same transaction/customer/card/merchant structure as the analytical layer, keeping the dashboard traceable to the Excel work.

Audit item	Observed result	Assessment
Transaction records	100,000	Pass
Customer records	25,000	Pass
Exact duplicate records	None identified in the completed analysis	Pass
Key fields used in KPIs	Available for analysis	Pass
Analytical calculated fields	Transaction Volume, Average Transaction Value, Transactions per Customer, Success Rate, Fraud Rate	Used consistently

Quality conclusion: the dataset supports the planned descriptive analysis. Any limitations are reflected in the interpretation and recommendations rather than hidden.

5. Hypothesis Framework

The hypotheses are framed around the main business questions and tested later using the completed descriptive analysis. Each hypothesis links a driver variable to a measurable outcome.

ID	Business question / hypothesis	Driver	Outcome	Expected direction
H1	Customers owning more cards have higher transaction engagement.	Card ownership	Avg. transactions / customer	Positive
H2	Higher-income customers have higher transaction values.	Income quartile	Avg. transaction value	Positive
H3	International transactions have materially different value and risk characteristics.	International flag	Avg. value + fraud rate	Higher
H4	Higher merchant risk classifications are associated with higher observed fraud rates.	Merchant risk level	Fraud rate	Positive
H5	Customer segments materially differ in transaction engagement.	Customer segment	Avg. transactions / customer	Positive

Why this structure matters: it creates a direct chain from business question -> driver -> outcome -> evidence -> decision, instead of treating the project as a list of disconnected charts.

6. Hypothesis Validation

The project uses descriptive evidence to classify each hypothesis as Supported, Strongly Supported or Not Supported. These labels describe the strength of the observed pattern and do not imply statistical significance or causality.

ID	Evidence	Result	Business interpretation
H1	Avg. transactions per customer increase from 3.07 -> 6.22 -> 9.09 for 1, 2 and 3 cards.	Supported	Multi-card customers are a strong engagement segment and a logical cross-sell target.
H2	Avg. transaction value rises from ₹13,988.68 in Q1 to ₹17,350.39 in Q4.	Supported	Income is more useful for value targeting than for predicting transaction frequency.
H3	International avg. value ₹37,022.80 vs ₹15,333.03 domestic; fraud 19.15% vs 4.65%.	Strongly Supported	Small volume, but materially higher value and risk exposure.
H4	Fraud rates are 5.40% High, 5.38% Low and 5.36% Medium.	Not Supported	Current merchant risk labels do not materially discriminate observed fraud.
H5	Avg. transactions per customer remain about 3.92-4.04 across customer segments.	Not Supported	Customer segment labels alone do not create a meaningful engagement split.

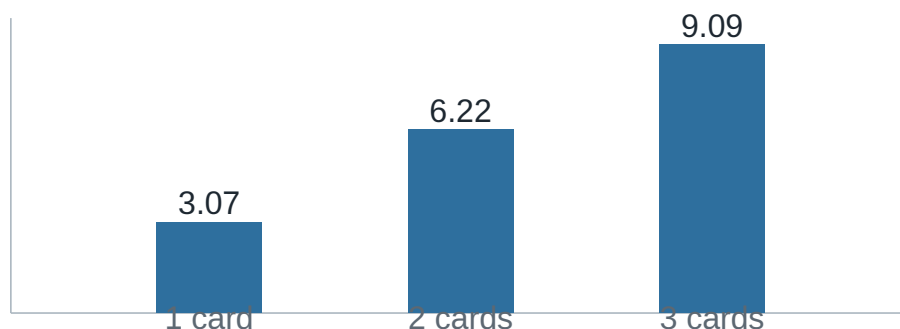
Validation summary: H1, H2 and especially H3 provide strong decision signals. H4 and H5 are useful mainly because they show where the business should not over-invest based on weak differentiation.

7. Customer Behaviour and Engagement EDA

Card ownership

Customers with more cards transact much more frequently. The relationship is strong and monotonic: 3.07 -> 6.22 -> 9.09 average transactions per customer across one, two and three cards. This is the strongest customer-level engagement signal in the project.

Average transactions per customer by card ownership



Customer segment and age

Customer segment averages remain tightly grouped around 3.92-4.04 transactions per customer. Age also shows limited differentiation, with average transactions spanning roughly 3.93-4.08 across the six age groups. These variables are better treated as supporting context than as primary engagement levers.

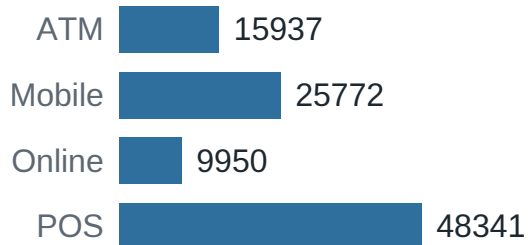
Income

Average transaction value increases from ₹13,988.68 in the lowest income quartile to ₹17,350.39 in the highest. The key distinction is that income is more clearly associated with transaction value than with transaction frequency.

8. Transaction Performance EDA

Channel performance

Transaction volume by channel



Success rate by channel



POS is the dominant channel with 48,341 transactions, followed by Mobile App at 25,772. Online Web has the lowest volume at 9,950. Despite the difference in volume, success rates are tightly grouped: 90.73%-91.24%.

Payment method mix

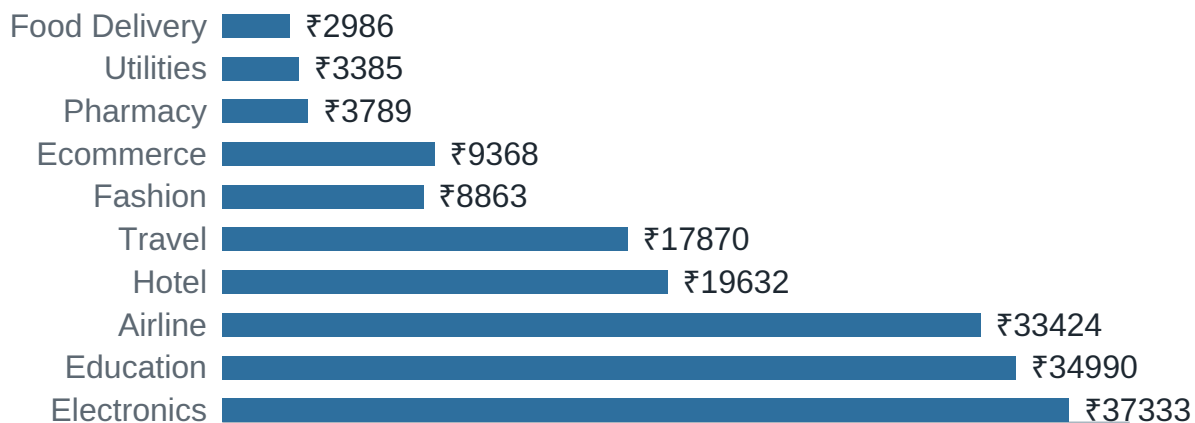
Payment method	Transactions	Share of total
Debit Card	51,217	51.2%
Credit Card	35,380	35.4%
UPI	10,971	11.0%
Net Banking	2,432	2.4%

Debit Card is the dominant payment method. The mix is useful operationally, but merchant category and international status create stronger strategic differentiation in the current analysis.

9. Merchant and Transaction Value EDA

Merchant category

Average transaction value by selected merchant categories

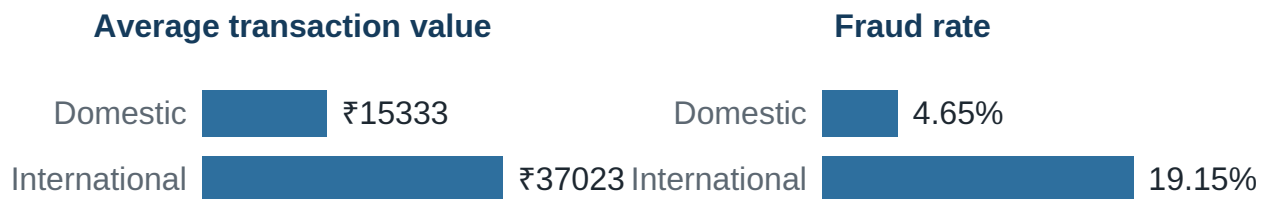


Merchant category shows meaningful variation in average transaction value. The highest observed values are Electronics (₹37,333.23), Education (₹34,989.99), Hospital (₹35,468.87) and Airline (₹33,424.42). Food Delivery, Utilities and Pharmacy sit at the lower end.

Business implication: merchant category can support differentiated offers, value segmentation and category-level monitoring. Treating all merchant activity the same would hide meaningful differences in transaction value.

10. Risk and International Transactions

Domestic vs International



International transactions are only about 5.0% of total transaction volume (5,033 of 100,000), but the average transaction value is more than twice the domestic level and the fraud rate is more than four times as high.

This is the clearest risk-prioritisation signal in the project. The business should treat international activity as a focused monitoring segment rather than applying broad controls across the full transaction base.

Merchant risk classification

Merchant risk level	Transactions	Fraud transactions	Fraud rate
High	12,391	669	5.40%
Low	61,947	3,334	5.38%
Medium	25,662	1,376	5.36%

The almost identical fraud rates show that the existing labels do not materially differentiate observed fraud. The labels can remain useful as one input, but should be supplemented with transaction-level and behavioural signals.

11. Merchant Geography and Tableau Design

Merchant geography

State	Average transaction value
West Bengal	₹20,125.77
Tamil Nadu	₹19,037.82
Uttar Pradesh	₹19,037.70
Telangana	₹17,508.97
Rajasthan	₹17,348.69
Kerala	₹17,673.59

Merchant geography adds useful context for value targeting. The analysis shows meaningful variation across states, with West Bengal, Tamil Nadu and Uttar Pradesh among the higher-value states in the completed analysis.

Stakeholder-focused Tableau views

View	Purpose	Core visuals	Relevant filters
Executive Overview	Portfolio baseline and operating health	KPI cards, channel volume, channel success	Transaction Type, Customer Segment
Customer & Value	Explain engagement and transaction-value differences	Card ownership, merchant category value, income view	Customer Segment, Income Quartile
Risk & Transactions	Focus management attention on high-value and high-risk activity	International value, international fraud, merchant-risk fraud	Transaction Type, Merchant Risk Level

Dashboard principle: use a small number of meaningful filters that answer real stakeholder questions. The Tableau build should remain connected to the same underlying fields used in the Excel analysis.

12. Recommendations and Expected Business Impact

Recommendation	Action	Expected business impact	Priority
Deepen multi-card engagement	Target customers with 2-3 cards using cross-sell, bundled benefits and relationship-led engagement.	Higher transaction frequency and stronger customer value.	High
Strengthen international controls	Use differentiated review and monitoring rules for international and high-value cross-border activity.	Lower fraud losses while protecting legitimate value.	High
Improve fraud detection beyond merchant risk	Combine merchant risk with amount, channel, device, payment method, international status and behavioural signals.	Better fraud detection and prioritisation.	High
Protect high-volume channels	Prioritise reliability monitoring for POS and other major transaction touchpoints.	Reduced friction at the highest-volume channels.	High
Use merchant category for value targeting	Create category-level offers and monitoring for high-value spending contexts.	Better targeting and stronger category-level understanding.	Medium
Use income selectively	Use income as a supporting signal for premium/value propositions rather than as an engagement predictor.	More focused high-value customer propositions.	Medium

Implementation sequence

Immediate: multi-card engagement, international monitoring, fraud-signal enhancement and POS reliability. **Medium term:** category-level targeting and income-informed value propositions. **Ongoing:** quarterly KPI monitoring and hypothesis refresh.

13. BRD-Style Decision Summary

Business problem

The bank has a large and active customer base, but not every customer or transaction dimension explains engagement, value and risk equally. Management needs a focused view of the few variables that deserve action.

Decision area	Evidence	Decision
Customer engagement	3.07 -> 6.22 -> 9.09 transactions per customer by card ownership	Prioritise multi-card engagement and cross-sell.
Value targeting	Higher income quartiles and merchant categories show higher transaction values	Use income and category as supporting value signals.
Risk management	International fraud 19.15% vs 4.65% domestic	Apply targeted international monitoring.
Fraud model quality	Merchant risk fraud rates are 5.36%-5.40%	Add richer behavioural and transaction-level signals.
Operations	POS = 48,341 transactions; channel success is about 90.7%-91.2%	Protect reliability where volume is concentrated.

Suggested success measures after implementation

Initiative	Primary KPI	Direction to monitor
Multi-card engagement	Avg. transactions per customer	Increase for the target segment
International controls	International fraud rate	Decrease without excessive false positives
Fraud-signal enhancement	Fraud rate separation across risk bands	Improve separation between risk groups
Channel reliability	Success rate by channel	Maintain or improve, especially for high-volume channels
Category targeting	Avg. transaction value	Increase value in priority categories

14. Conclusion, Limitations and Next Steps

Conclusion

The analysis identifies a small set of strong and practical signals. Multi-card ownership is strongly associated with higher engagement. International activity combines high value with materially higher fraud exposure. Merchant category creates meaningful differences in transaction value. In contrast, customer segment, age and merchant risk level show limited separation in the observed data.

The recommended strategy is therefore selective: deepen engagement where behaviour already signals stronger usage, use income and merchant category to support value targeting, protect high-value international activity, and strengthen fraud controls with richer transaction-level signals.

Limitations

- The analysis is descriptive and should not be interpreted as causal.
- Additional time periods would improve stability and enable stronger trend analysis.
- External market benchmarks were not part of the available project scope.
- Operating-cost and profitability measures are outside the current transaction-focused analysis.

Next steps

- Track the core KPIs monthly or quarterly and refresh the hypotheses as new data arrives.
- Pilot multi-card engagement interventions and measure changes in transaction frequency.
- Build a targeted international monitoring approach and measure fraud reduction.
- Enrich the fraud framework with behavioural, device, channel and transaction-value signals.
- Use Tableau as the recurring stakeholder monitoring layer and Excel as the traceable analytical workbook.

Final outcome: the project provides an evidence-led view of customer engagement, transaction value, operational performance and fraud exposure. The main business opportunity is not to optimise every variable equally, but to focus effort on the dimensions that show the clearest practical separation in the data.